
CS 104.3 -IT Solution Architecture

Lesson 1-Introduction to Computer Architecture

About Module

- Module Code: CS 104.3
- Module Title: IT Solution Architecture
- Level: 3
- Assignment
 - Assignment : 40%
 - Final Exam : 60%

Learning Outcomes

1. Explain the fundamental principles of CPU architecture, including core components and performance considerations.
2. Describe the role of Instruction Set Architecture (ISA) and its relationship between hardware and software systems.
3. Analyze memory hierarchies and storage architectures in terms of performance, scalability, and system requirements.
4. Explain the importance of abstraction, virtualization, and system interfacing in modern computing environments.
5. Apply basic system and solution architecture principles to design simple, scalable IT solutions at a conceptual level.

References

- *Computer Organization and Design MIPS Edition, Edition 6* *The Hardware/Software Interface* by David A. Patterson and John L. Hennessy (2020)
- *Fundamentals of Software Architecture: An Engineering Approach* 1st Edition by Mark Richards, Neal Ford (2020)



Who is this?



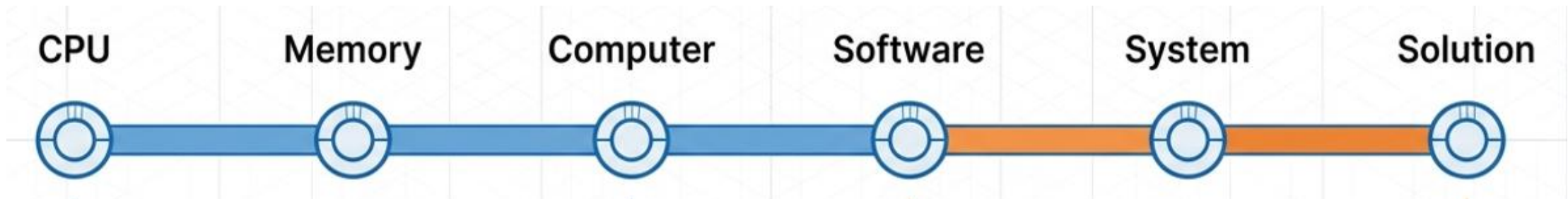
Who is this?



Who is this?

What will be discussed?

Architecture in computing is the high-level design, structure, and organization of a system—describing its components, their relationships, and how they work together to achieve a specific goal.

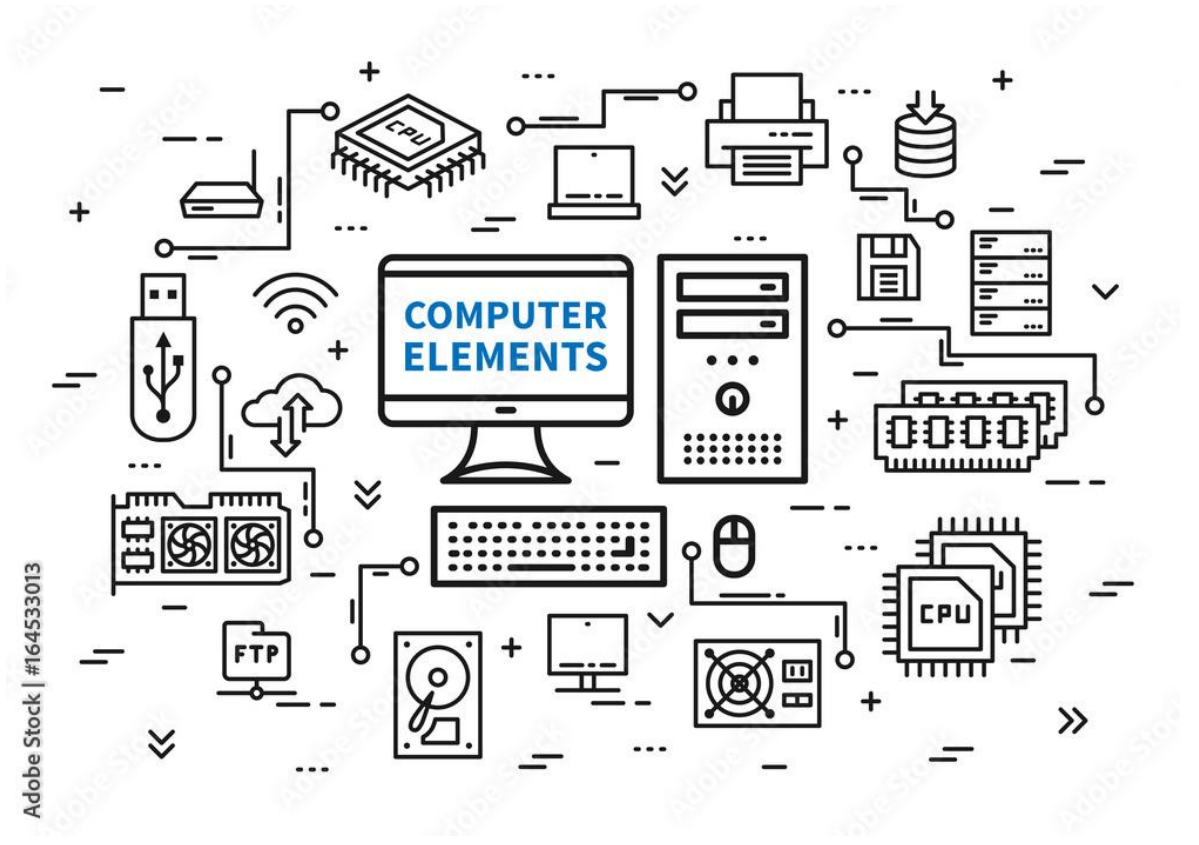


Answer

- What's inside your computer?
 - Software Vs Hardware?
 - Input Devices?
 - Output Devices?
 - Types of Computers?

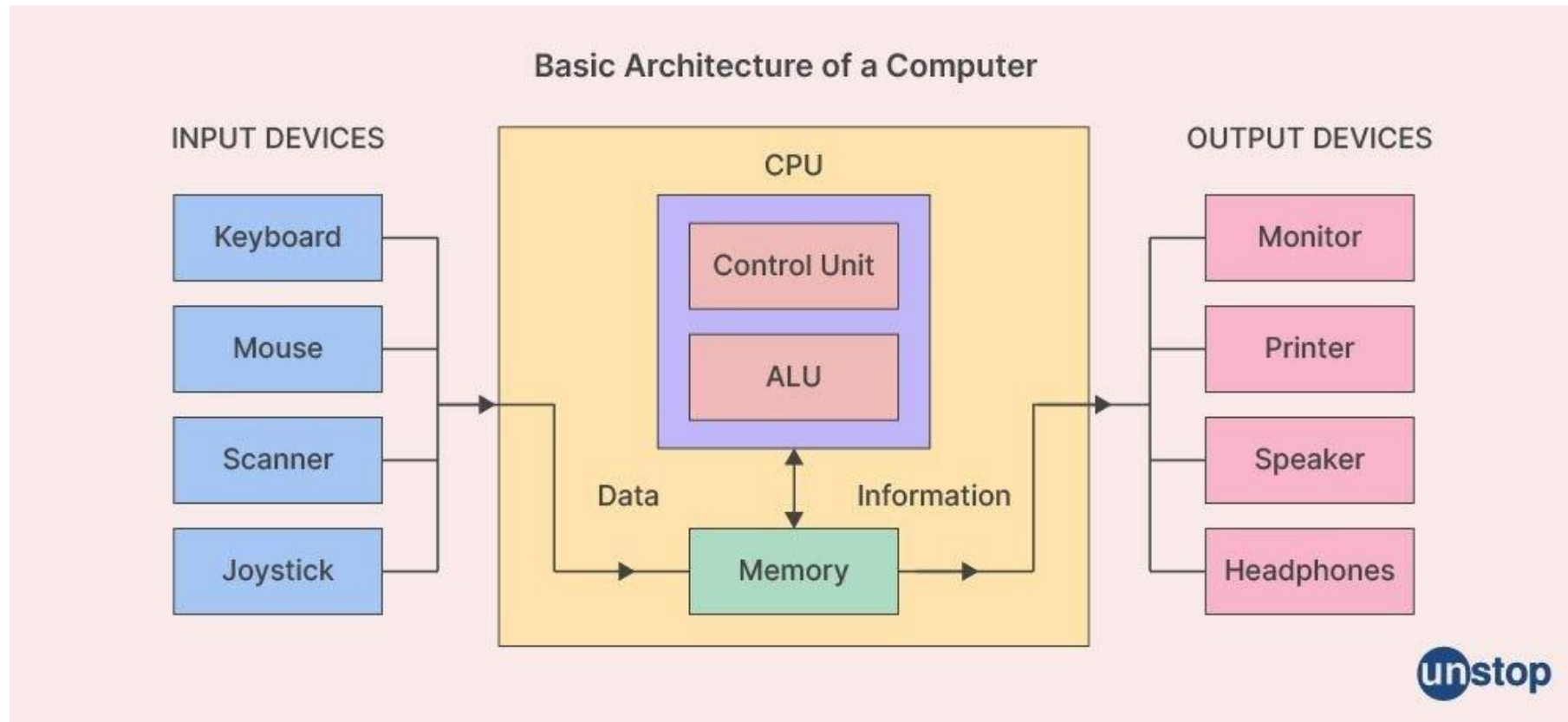


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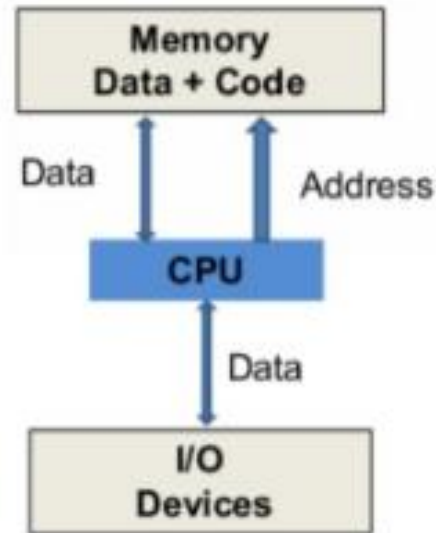
[How CPU works explained in 60 seconds | TheMVP](#)

Computer Architecture

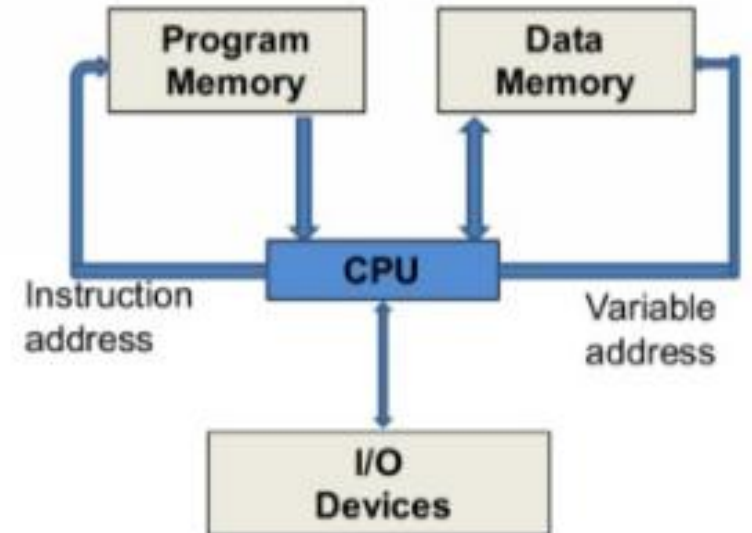


Von Neumann vs Harvard

- Von Neumann
 - Single memory for both instructions and data.
- Harvard
 - Separate memory for instructions and data.

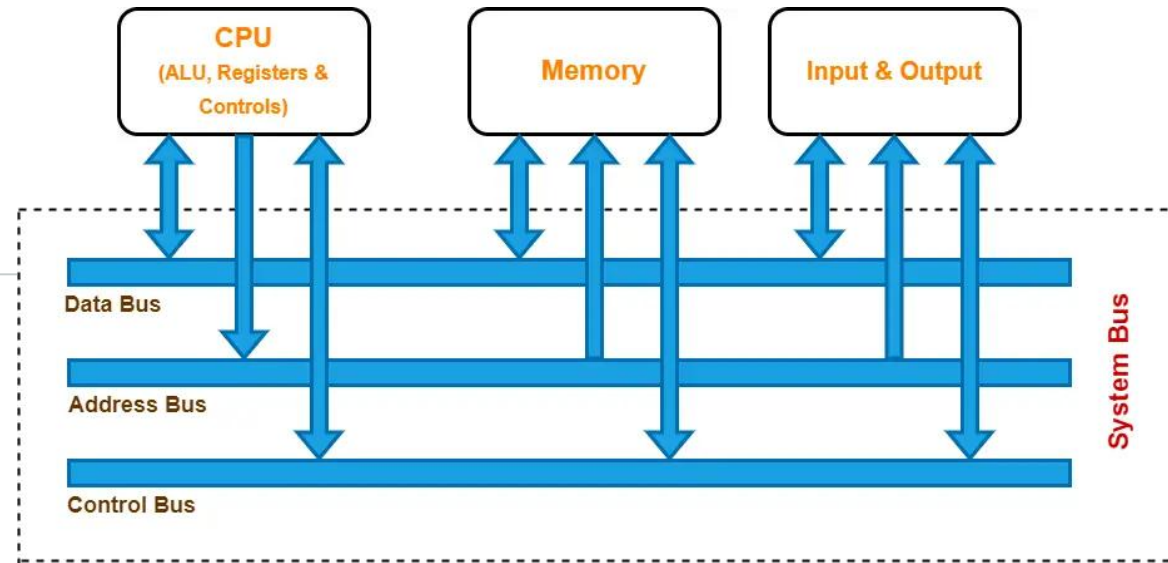


Von Neumann Machine



Harvard Machine

System buses

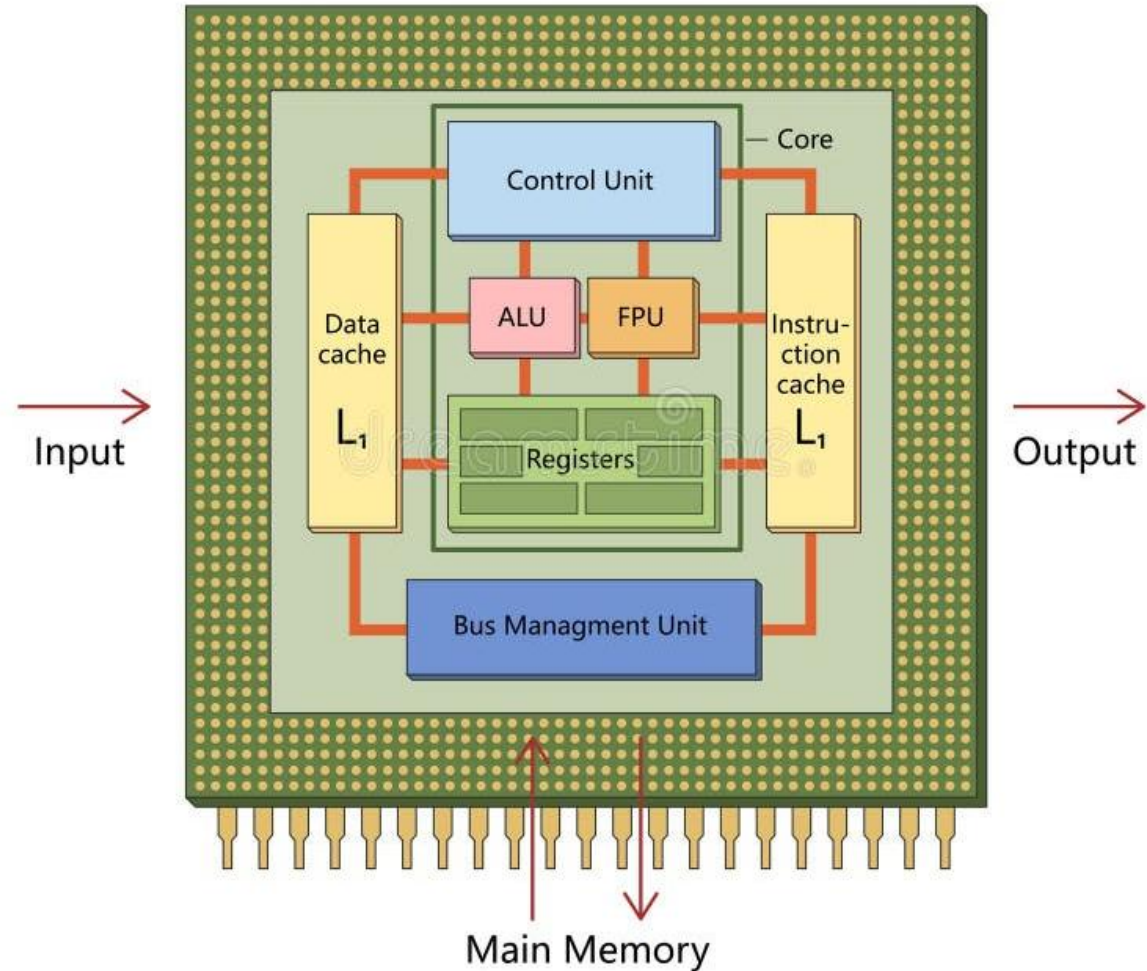


Bus Type	Purpose	Direction	Example
Address Bus	Carries the address(Location) of the data	CPU → Memory/I/O	CPU sends address 1000
Data Bus	Carries the actual data	Both directions	Memory sends data to CPU
Control Bus	Carries control signals	Both directions	READ, WRITE, Interrupt

[How are CPUs made? - YouTube](#)

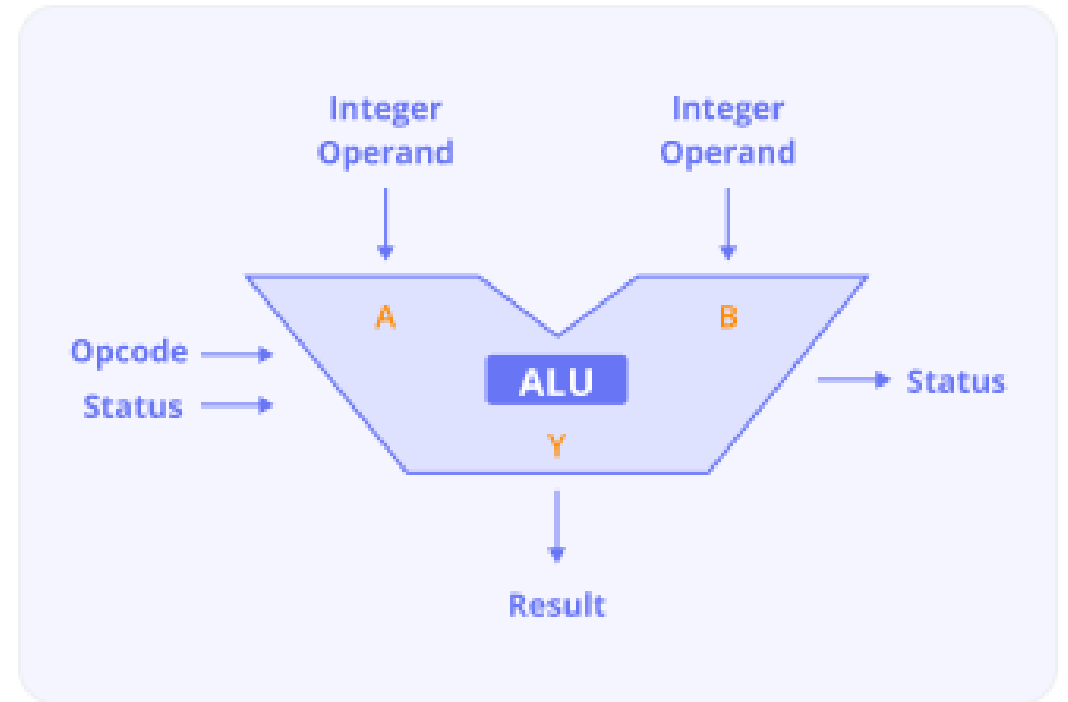
CPU Architecture

- **Arithmetic Logic Unit (ALU):** Performs calculations and logical operations.
- **Control Unit:** Directs operations by sending control signals.
- **Registers:** Small, fast storage locations inside the CPU.



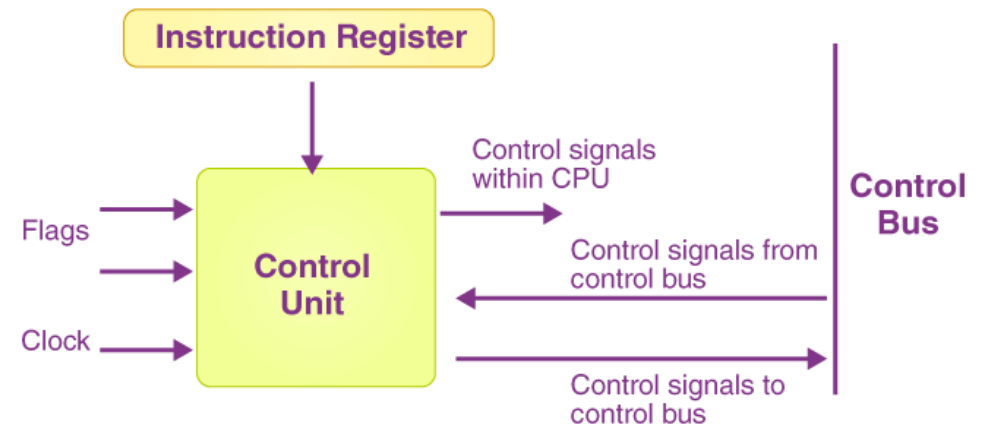
Arithmetic and Logic Unit

- Performs arithmetic operations (add, subtract, multiply, divide).
- Performs logic operations (AND, OR, NOT, comparisons).
- Example:
 - If you type $2 + 3$, the ALU adds them.



Control Unit

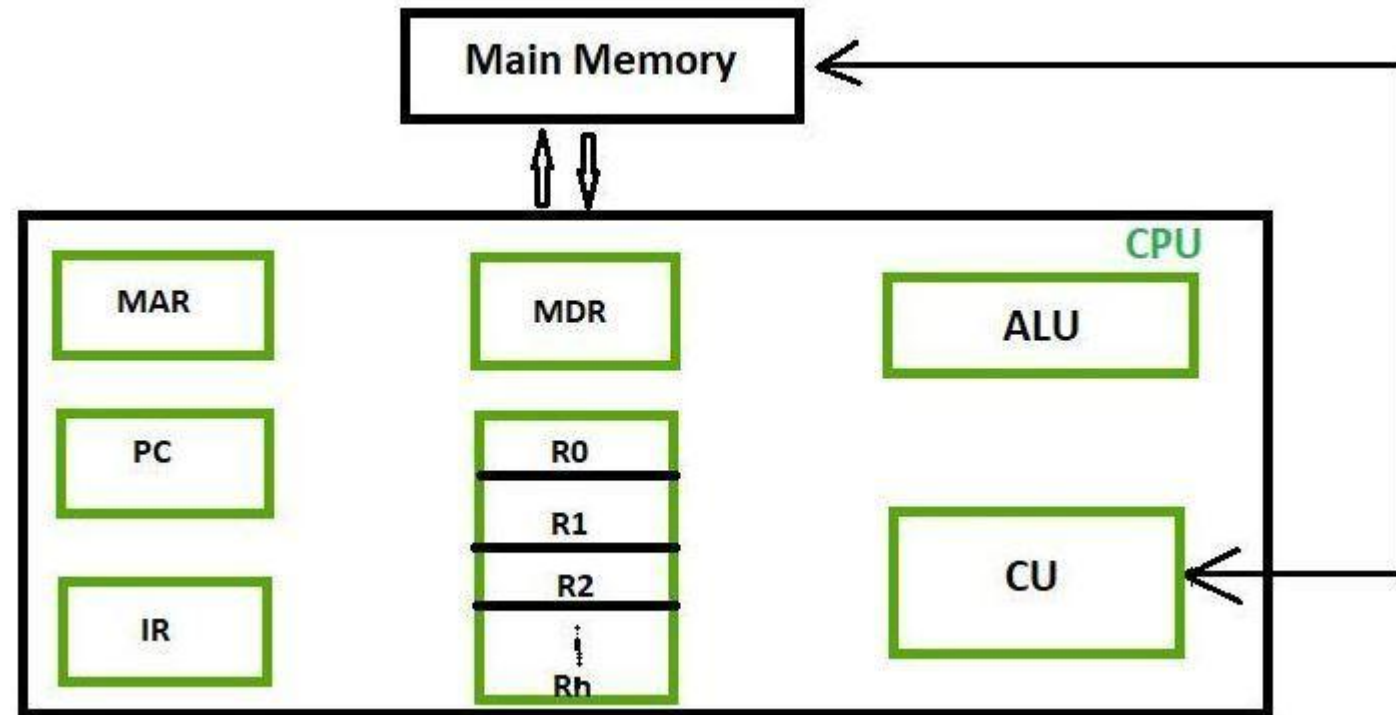
- Directs everything the computer does.
- Tells memory, ALU, and input/output devices what to do.
- Sends control signals to coordinate all parts.



Block Diagram of the Control Unit

Registers

- Small, very fast storage areas inside the CPU.
- Hold data and instructions temporarily while being processed.



Quick Question

- Which part of the CPU performs calculations?
- Which part tells other components what to do

Activity

- Being in pairs (02) Search for different register types.

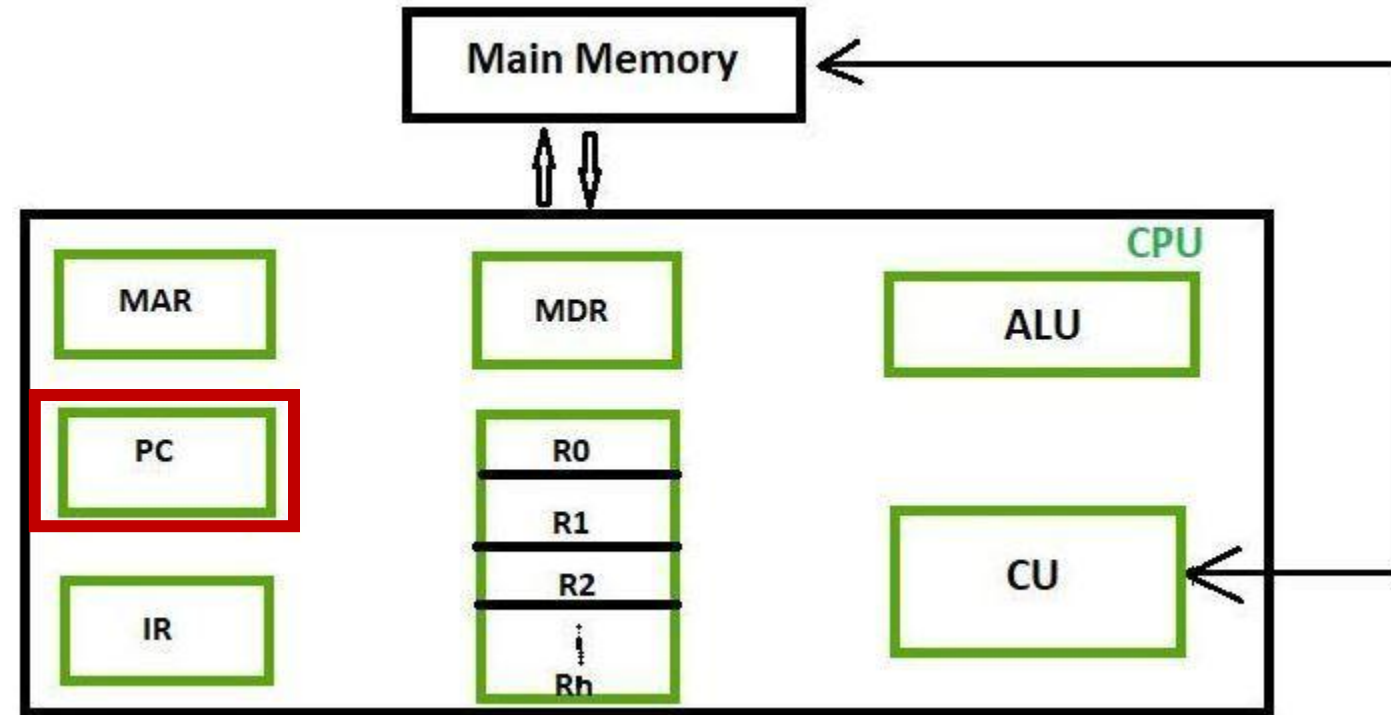
How Many Registers in CPU

- It depends on the **CPU architecture**.
- Simplified / Educational Models Used for teaching (like in textbooks or simulators like MARIE)
 - Typically: **4–8 registers** in these models.

Register	Purpose
A or ACC	Accumulator (main register for arithmetic)
B	General-purpose register
PC	Program Counter
IR	Instruction Register
MAR	Memory Address Register
MDR	Memory Data Register
FLAGS	Status/Condition flags
SP	Stack Pointer

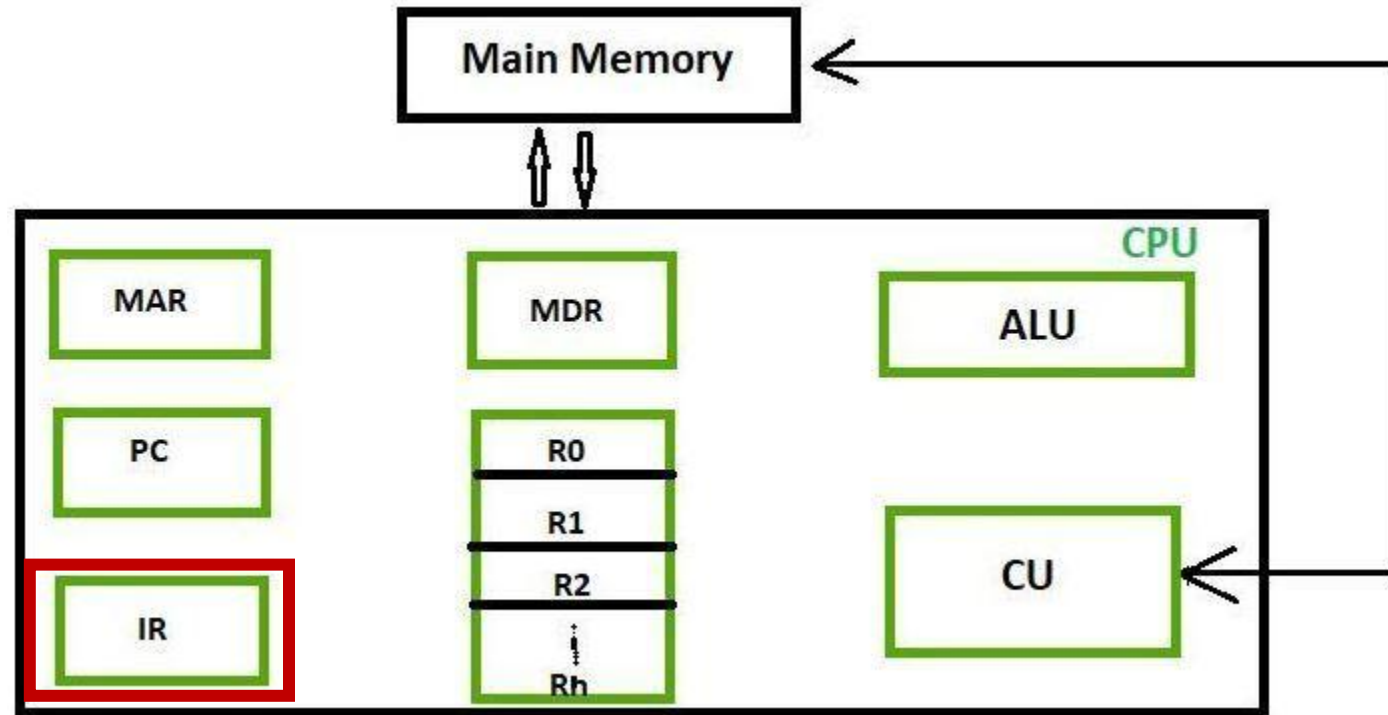
Program Counter(PC)

- Holds the **address** of the next instruction to fetch.
- After fetching, it increments to point to the next instruction.
- Tells the CPU *where* to get the instruction.



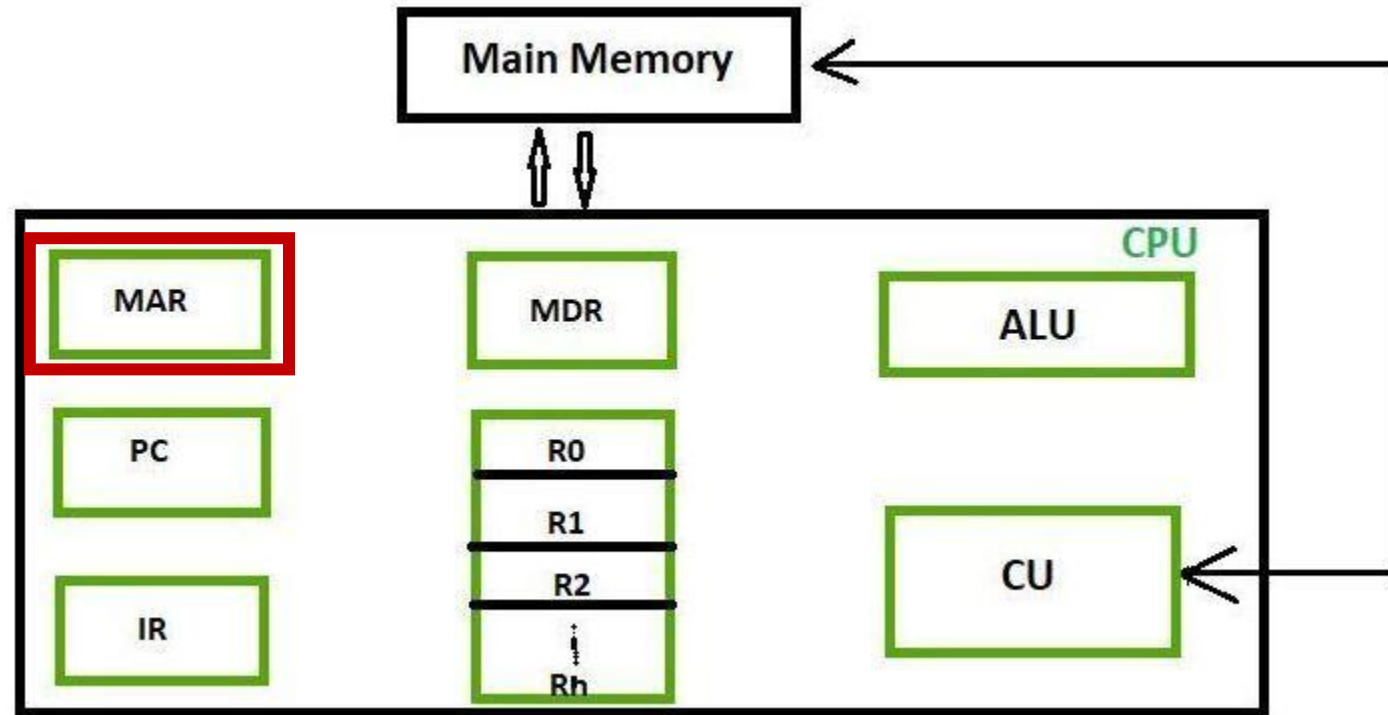
Instruction Register(IR)

- Holds the instruction currently being decoded and executed.
- Contains the instruction bits so the Control Unit can interpret them.



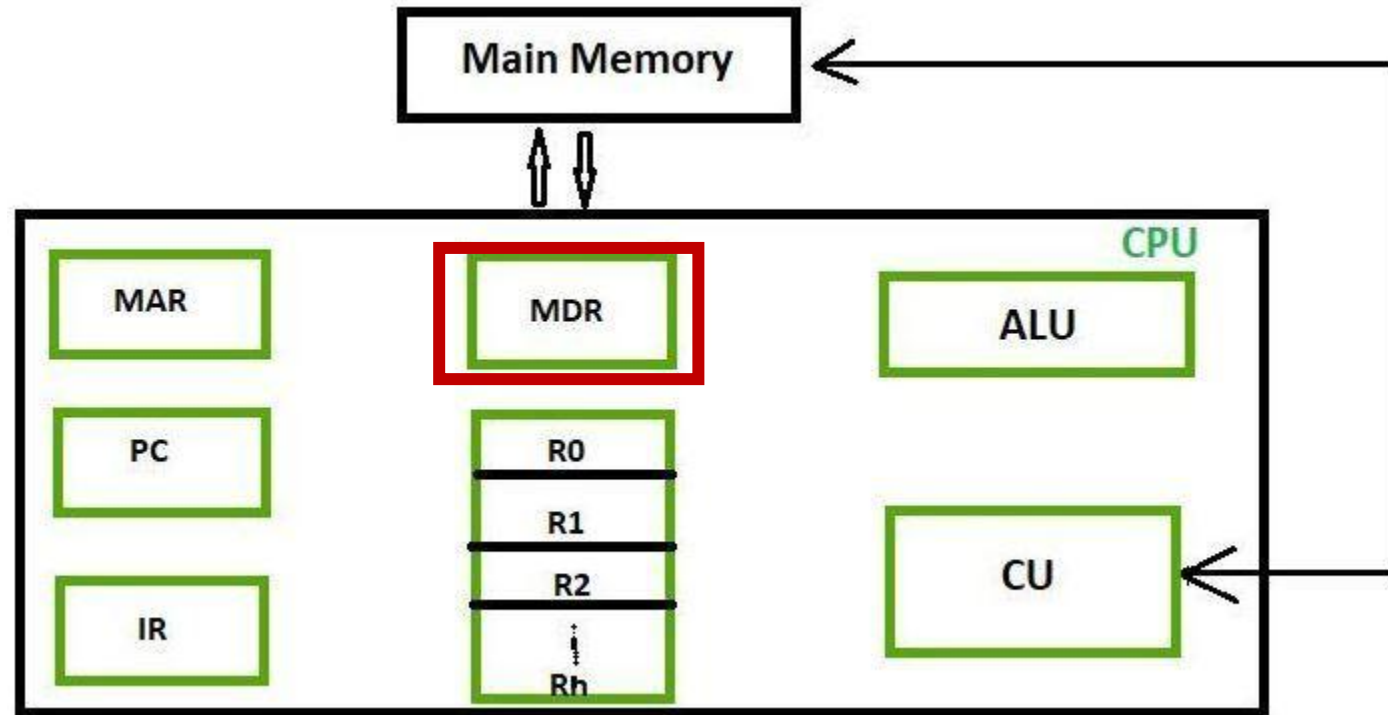
Memory Address Register (MAR)

- Holds the **address in memory** that the CPU wants to read or write.
- Gets the address from PC to fetch the instruction.



Memory Buffer Register (MBR)

- Also known Memory Data Register (MDR)
- Holds the **data** coming from memory or going to memory.
- Temporarily stores fetched instructions or data.



Accumulator (ACC)

- Holds intermediate arithmetic and logic results.
- Stores results produced by the ALU.
- **B Register**
 - The **B register** is a **temporary storage register** inside the CPU.
 - Its main purpose is to **hold data temporarily** during **arithmetic or logic operations**.
 - **A register (Accumulator)**: stores one operand or the result of an operation.
 - **B register**: stores the **second operand** for arithmetic/logic operations.

Quick Recap

- List down the main components of CPU and its purposes
- Mention three types of system buses

Thank You!
